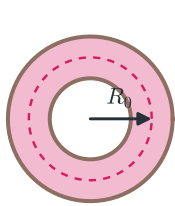
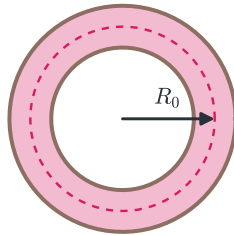


TWO WAYS TO GROW THE SAME MACHINE — and what each one exports

A. fixed wall loading W_n — the plant gets bigger because P_E did



reference



$R_0 \times 1.50$, wall area $\times 1.50$, $P_E \times 1.50$

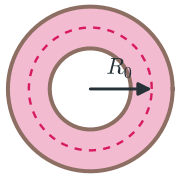
$$P_E \uparrow \Rightarrow R_0 a \uparrow \Rightarrow V \uparrow$$

$P_E \propto p^2 V$, and V grew by the same factor, so at fixed a :

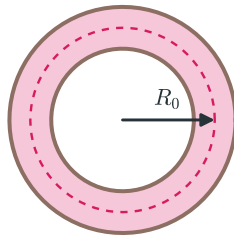
$$p \times 1.00 \text{ — UNCHANGED}$$

Exported: capital scale — nothing about the plasma itself got any easier.

B. fixed P_E , lower W_n — the plant gets bigger to spread the SAME neutron power



reference



$R_0 \times 1.50$, $W_n \times 0.67$, $p, n \times 0.82$

$$W_n \downarrow \Rightarrow V \uparrow \Rightarrow p, n \downarrow$$

same P_E over $\times 1.50$ the volume:

$$p \times 0.82, \quad W_n \times 0.67$$

Exported, and not small: MORE blanket, MORE shield, MORE TF-coil volume — all competing for the inboard radius \$46 already showed is full.

Both rows are the SAME drawing with a different quantity held fixed — which is why they are one figure, not two.

Every ratio is computed from $V = 2\pi^2 R_0 a^2$ and $A_w = 4\pi^2 R_0 a$ at fixed a ; none is estimated. Row A's "p unchanged" is an identity, not an approximation. A scaling sketch only: it says nothing about economics.

Explanatory sketch; schematic, not to scale. Plan view: the annulus is the first wall, its shading the pressure.

Not a cumulative machine state.